

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 12

Mobility Management Entity (MME)

موجودیت مدیریت تحرک (MME)



Lecture Notes

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1 Why MME Exists

The MME is the **central control-plane entity** in the LTE/EPC architecture. It exists to:

- **Offload the RAN from core signaling:** In 2G/3G, the RNC/BSC handled significant core-network signaling. LTE's flat architecture moved all NAS (Non-Access Stratum) signaling termination to the MME, simplifying eNodeB (eNB) design.
- **Centralize control-plane intelligence:** Authentication, mobility management, bearer setup, and paging are all orchestrated by the MME.
- **Decouple control and user planes:** The MME handles no user-plane traffic — it purely manages signaling, enabling independent scaling of CP and UP.

Key Interfaces

Interface	Peer Node	Protocol	Purpose
S1-MME	eNB	S1-AP (SCTP)	NAS transport, handover signaling
S6a	HSS	Diameter	Authentication vectors, subscription data
S11	S-GW	GTPv2-C	Bearer/session management
S10	Other MME	GTPv2-C	Inter-MME handover, context transfer
S3	SGSN	GTPv2-C	2G/3G ↔ LTE mobility

2 MME Functions

2.1 NAS Signaling Termination

- The MME is the NAS endpoint — it terminates all NAS messages from the UE (transported transparently through eNB via S1-AP).
- NAS messages are integrity-protected and ciphered between UE and MME.

2.2 Mobility Management

- Tracking Area (TA) management and TA List assignment
- Handover signaling (S1-based and X2 with MME involvement)
- Inter-RAT mobility (to/from 2G/3G via S3/S10)

2.3 Authentication and Security

- Retrieves authentication vectors from HSS (via S6a)
- Runs EPS-AKA (Authentication and Key Agreement)
- Derives NAS keys (K_{NASint} , K_{NASenc}) and AS keys (K_{eNB})
- Initiates NAS Security Mode Command

2.4 Bearer Management

- Establishes default EPS bearers during attach
- Coordinates dedicated bearer activation/modification/deactivation
- Interacts with S-GW (S11) and indirectly P-GW

2.5 Paging

- Pages UEs in ECM-IDLE state when downlink data arrives
- Sends S1-AP Paging messages to all eNBs in the UE's registered TA list

2.6 Idle-Mode Reachability

- Maintains UE context even when UE is IDLE (ECM-IDLE)
- Knows which Tracking Areas the UE may be in
- Triggers paging when data or signaling arrives

2.7 MME Selection, Pooling, and Rebalancing

- Multiple MMEs serve the same area (MME Pool)
- Load balancing via relative capacity (weight factor) advertised to eNBs
- Rebalancing: graceful transfer of UE contexts when adding/removing MMEs

3 NAS Protocol Stack

NAS (Non-Access Stratum) operates between UE and MME, transparent to eNB.

3.1 EMM – EPS Mobility Management

Manages UE registration and mobility:

- Attach / Detach
- Tracking Area Update (TAU)
- Authentication (EPS-AKA)
- Security Mode Control
- GUTI Reallocation
- Service Request

EMM States:

- EMM-DEREGISTERED: UE not registered
- EMM-REGISTERED: UE attached to network

3.2 ESM – EPS Session Management

Manages PDN connections and EPS bearers:

- PDN Connectivity Request/Accept
- Bearer Resource Allocation
- Bearer Modify / Deactivate
- PDN Disconnect

ESM operates piggybacked on EMM during attach (ESM message embedded in Attach Request).

4 Detailed Procedures

4.1 LTE Initial Attach (Full Flow)

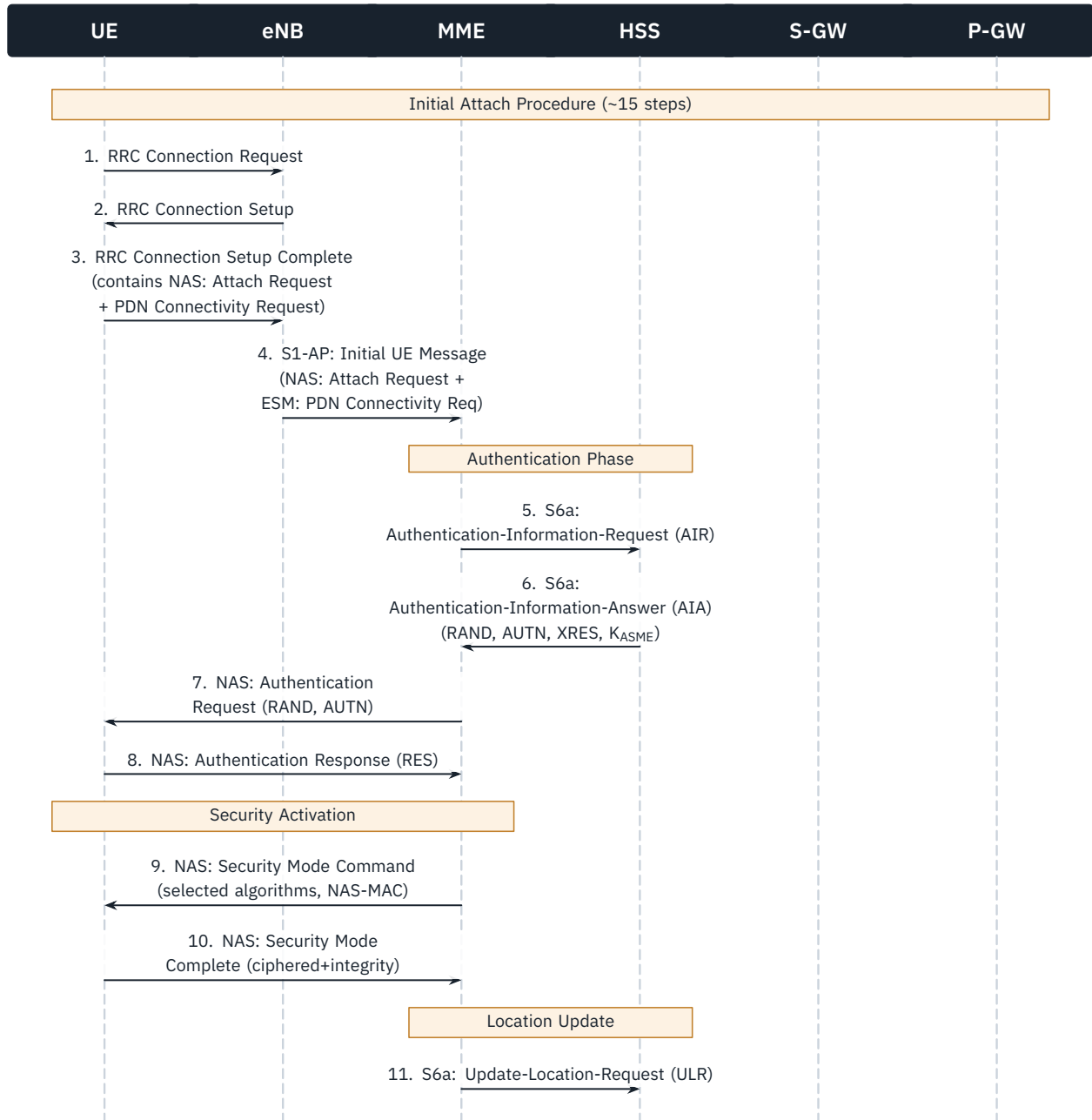


Figure 12.1 LTE initial attach (full flow) (part 1 of 2)

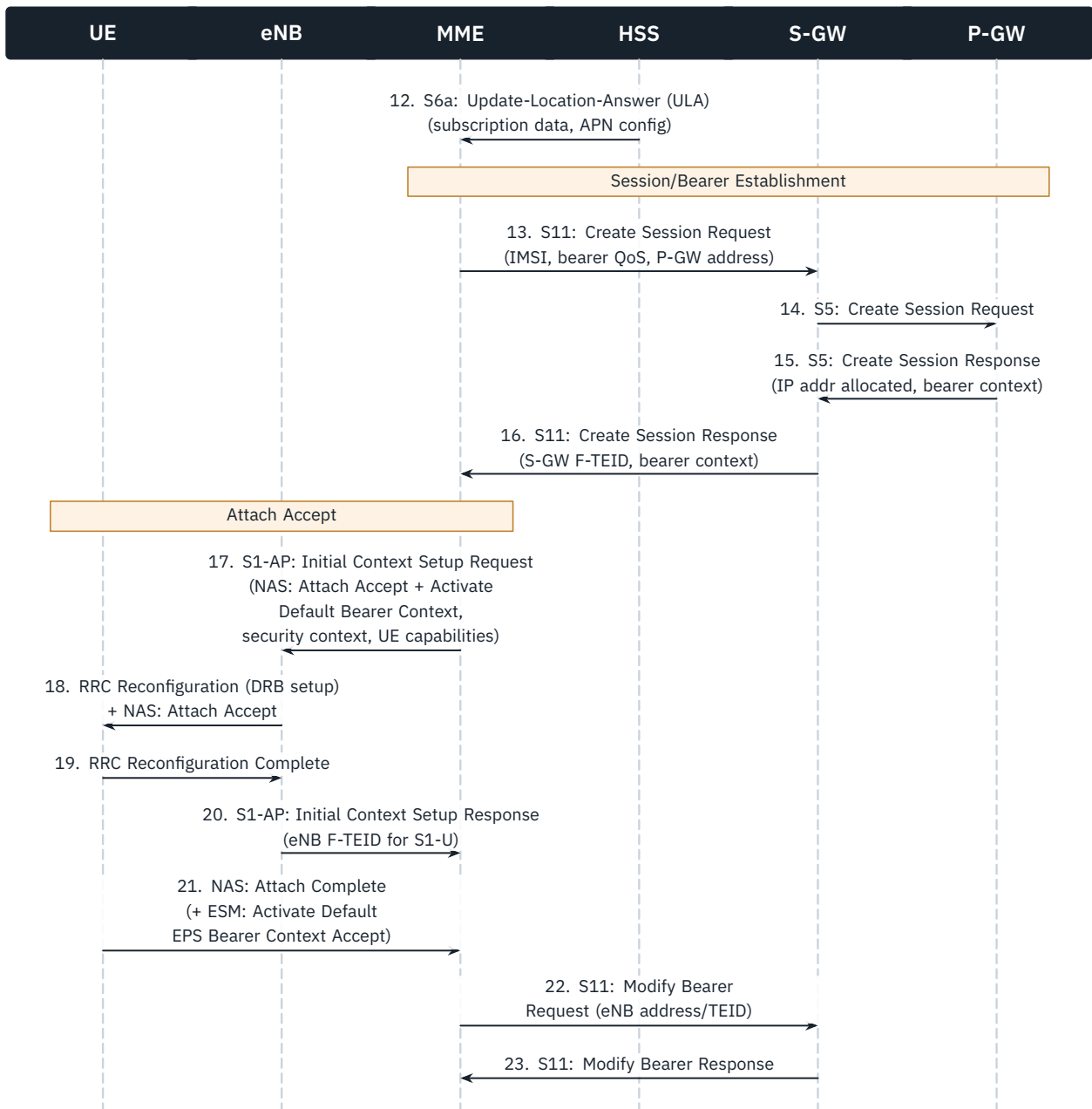


Figure 12.2 LTE initial attach (full flow) (part 2 of 2)

Key Points:

- Steps 7-10: EPS-AKA + NAS security activation
- Steps 13-16: GTP-C session establishment across S11/S5
- The Attach Request carries a piggybacked PDN Connectivity Request (ESM)
- The UE gets its IP address from P-GW (step 15)

4.2 Authentication and Security Mode

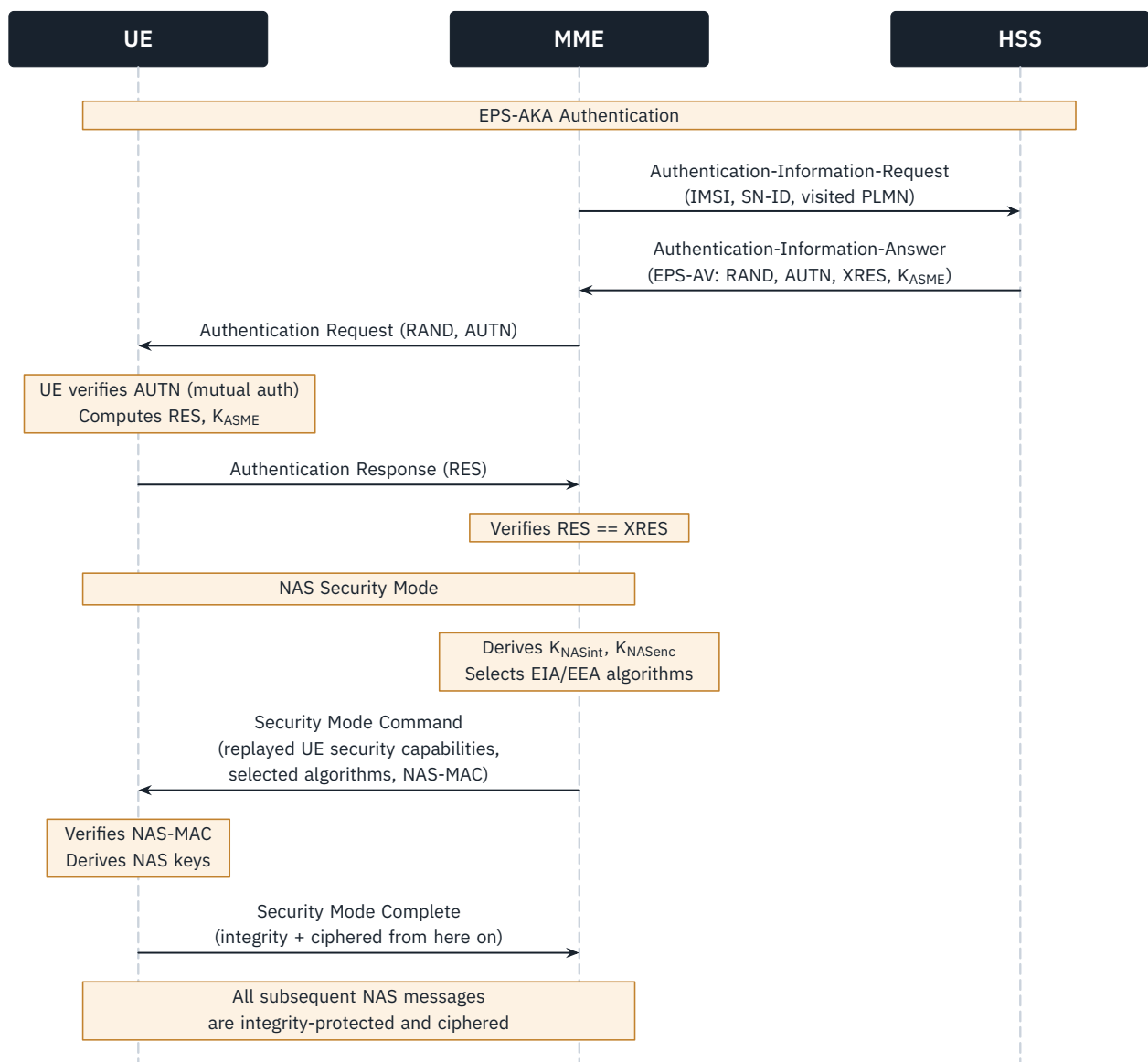


Figure 12.3 EPS-AKA authentication and NAS security mode

Key Derivation Hierarchy:

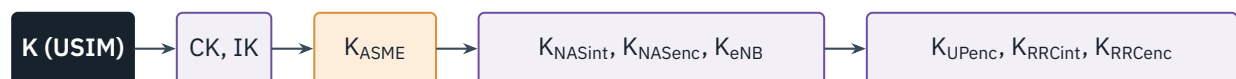


Figure 12.4 Key derivation hierarchy

4.3 Tracking Area Update (TAU)

Triggers:

- **Periodic TAU:** Timer T3412 expires (default ~54 min) — confirms UE is still reachable
- **Mobility-triggered TAU:** UE moves to a cell whose TAC is not in its TA List

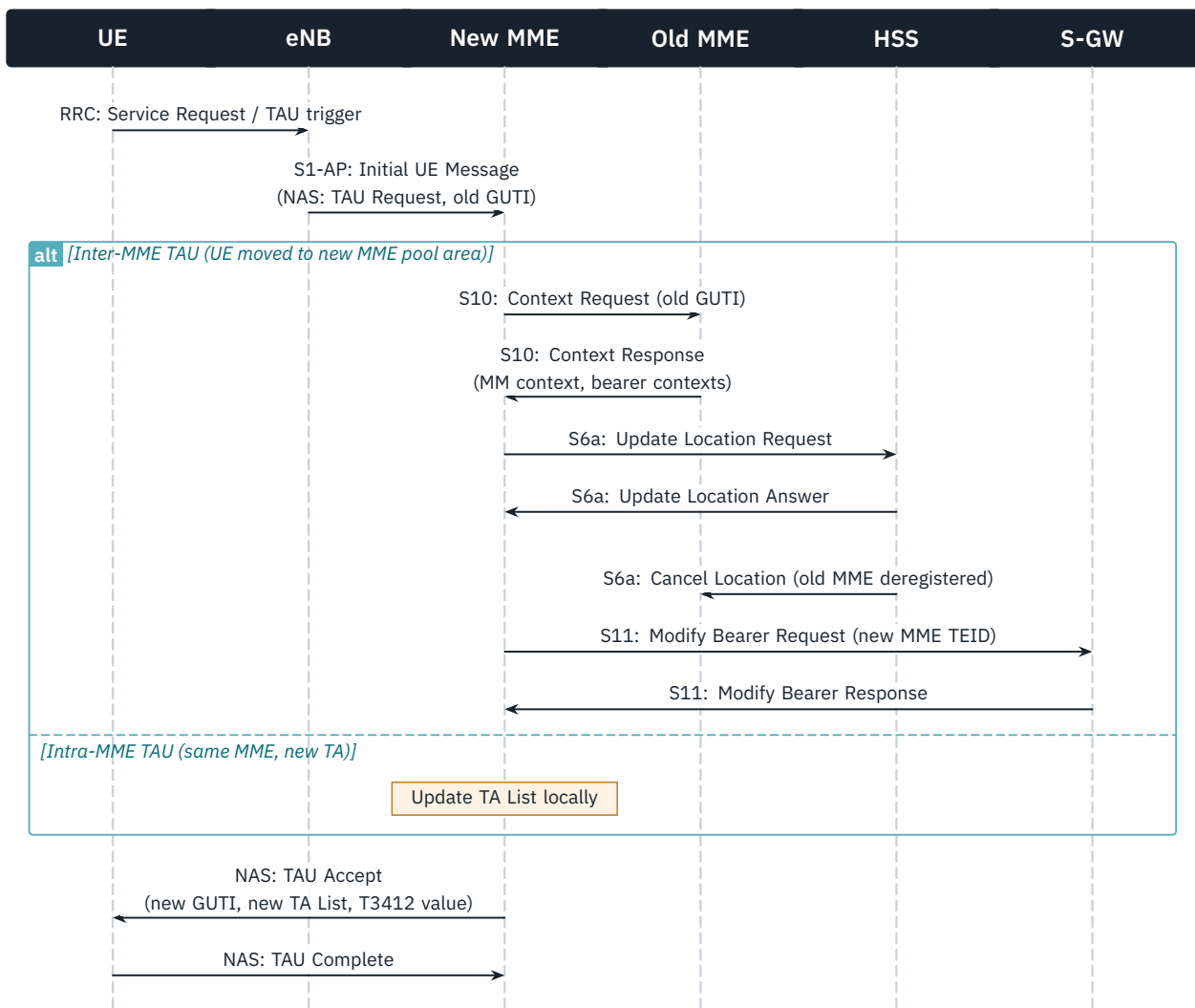


Figure 12.5 Tracking area update, inter-MME and intra-MME

4.4 Detach Procedure

■ UE-Initiated Detach

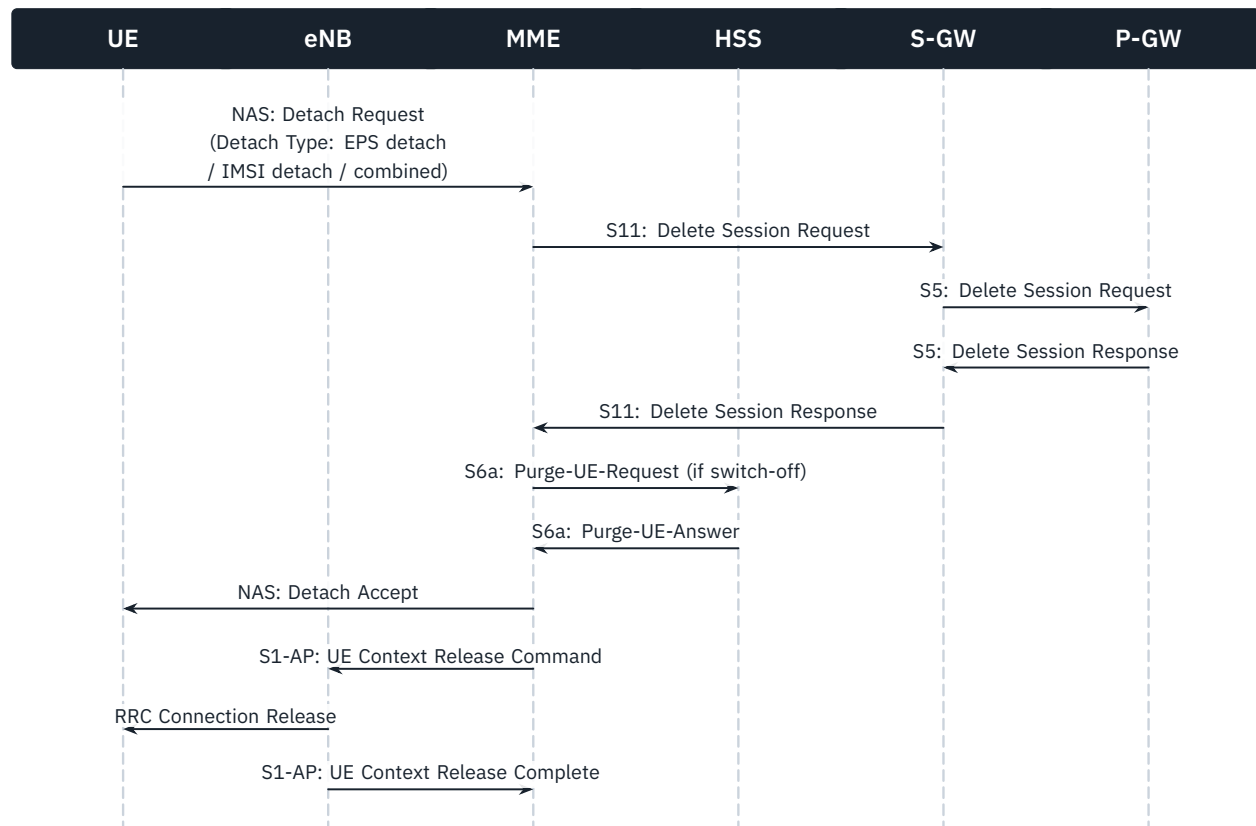


Figure 12.6 UE-initiated detach

■ Network-Initiated Detach

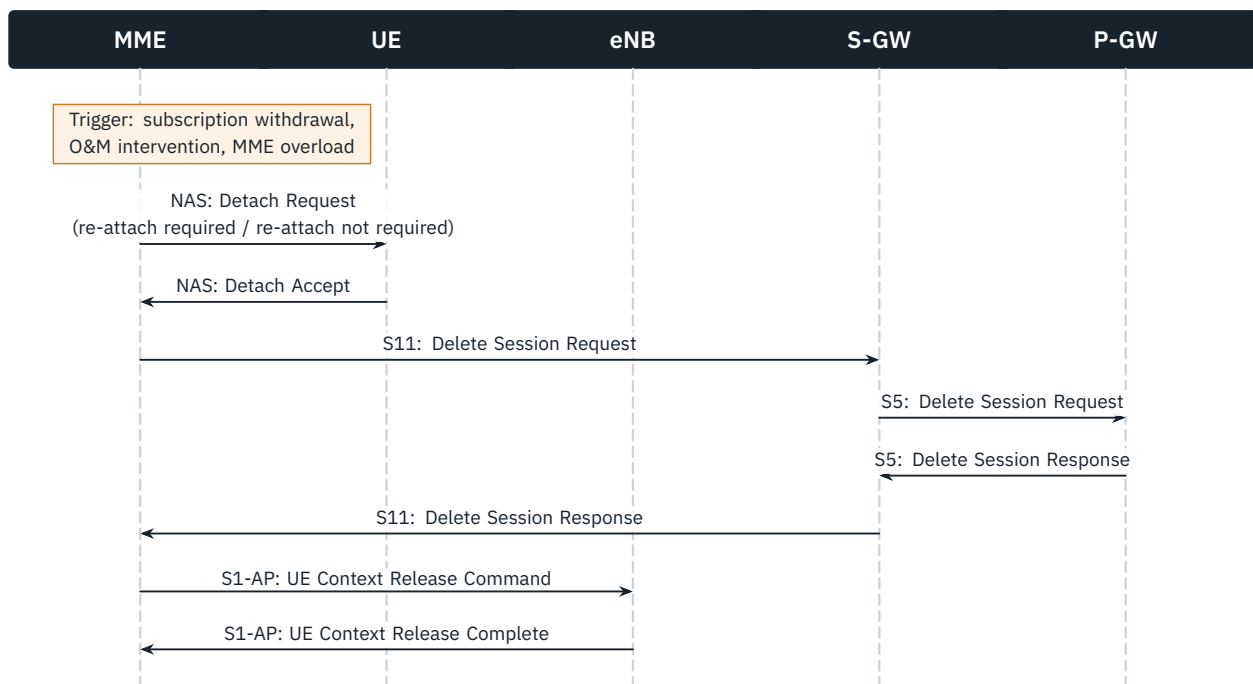


Figure 12.7 Network-initiated detach

4.5 Paging (S1 Paging Procedure)

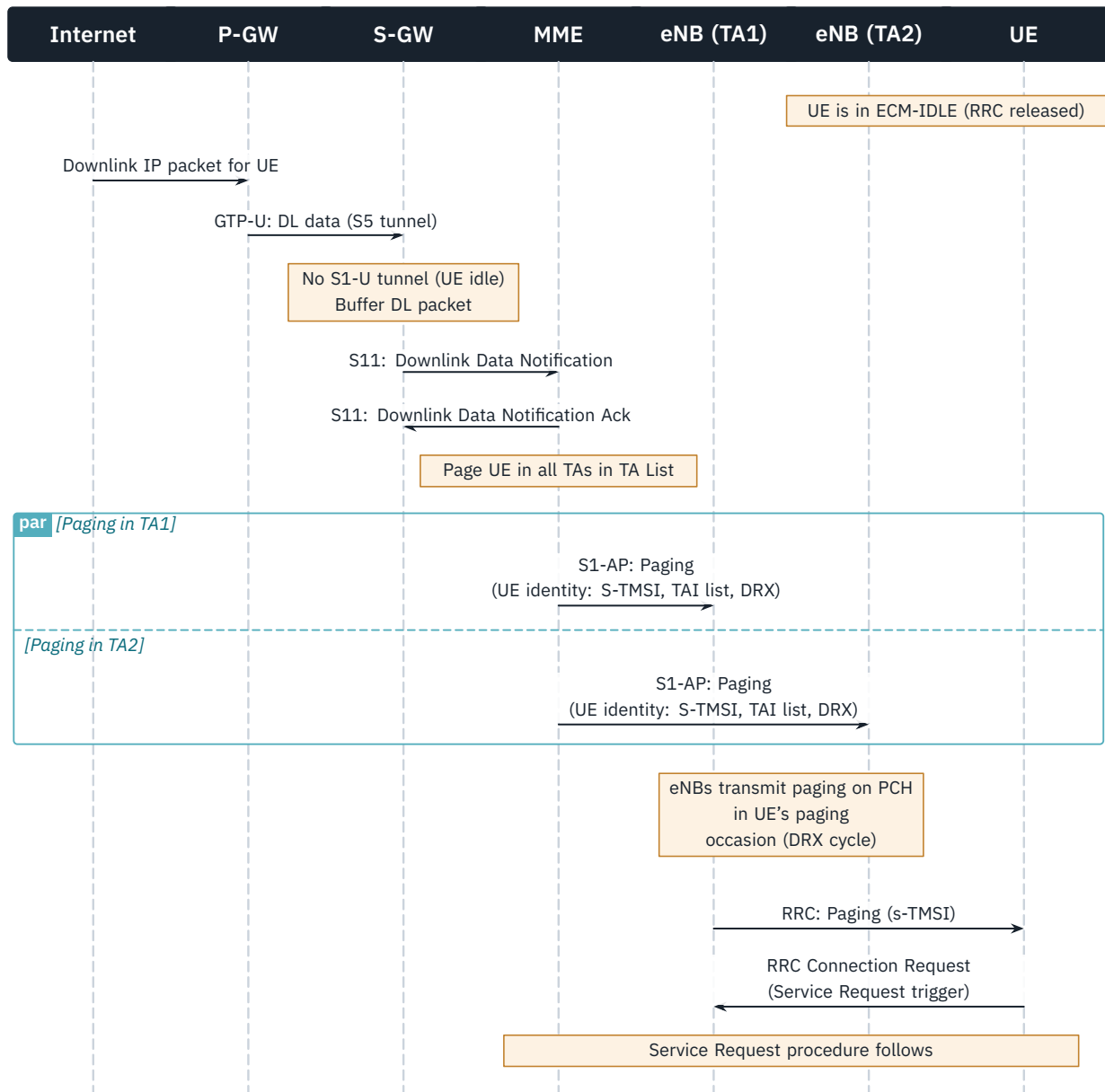


Figure 12.8 S1 paging of an idle UE

4.6 Service Request (UE in IDLE → CONNECTED)

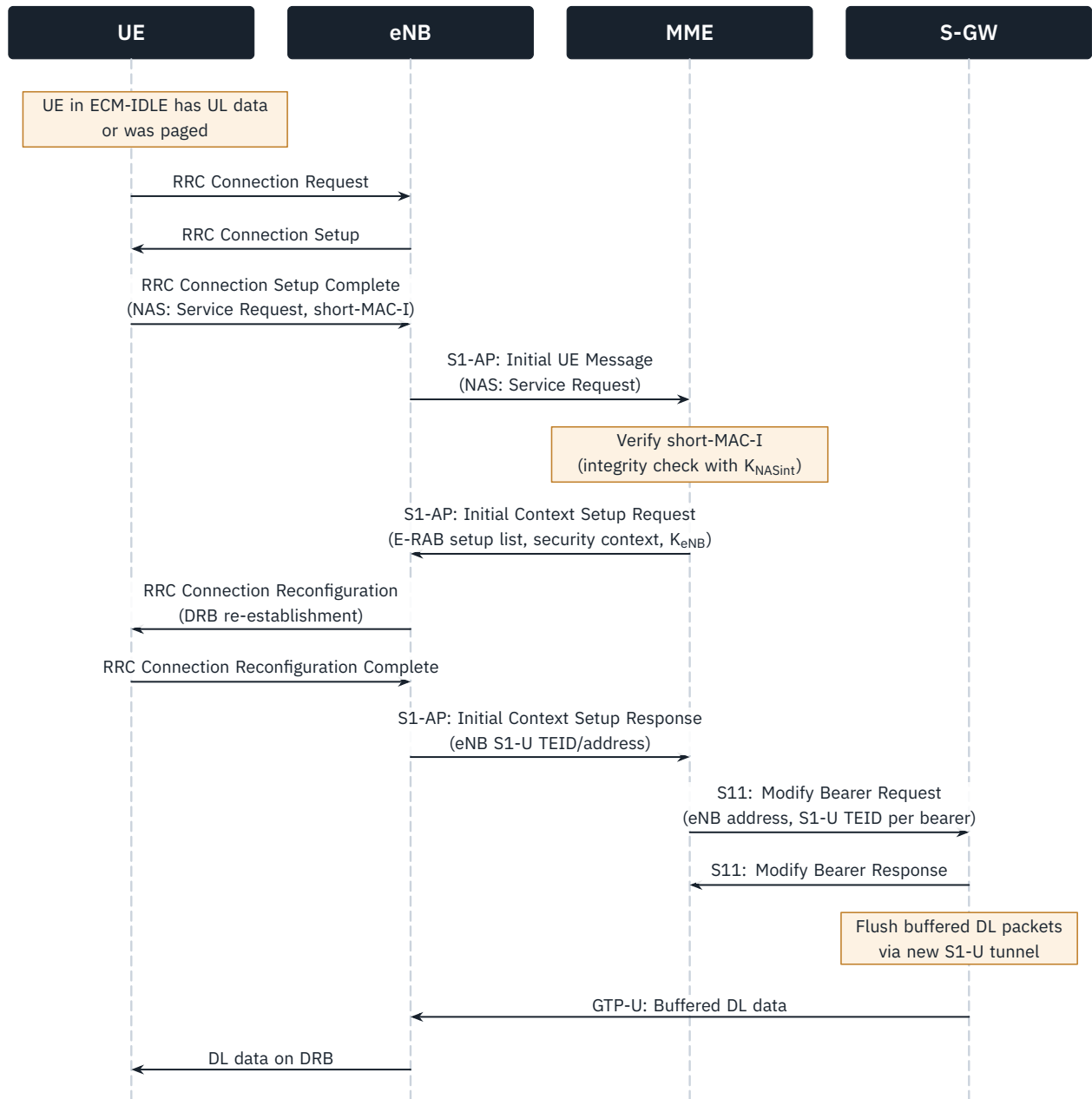


Figure 12.9 Service request (ECM-IDLE → ECM-CONNECTED)

5 Bearer Management

5.1 Default Bearer Setup

- Created automatically during Initial Attach (part of Create Session)
- One default bearer per PDN connection (always-on, non-GBR)
- QCI typically 9 (best-effort Internet) or 5 (IMS signaling)
- Cannot be deactivated without disconnecting the PDN

5.2 Dedicated Bearer Activation

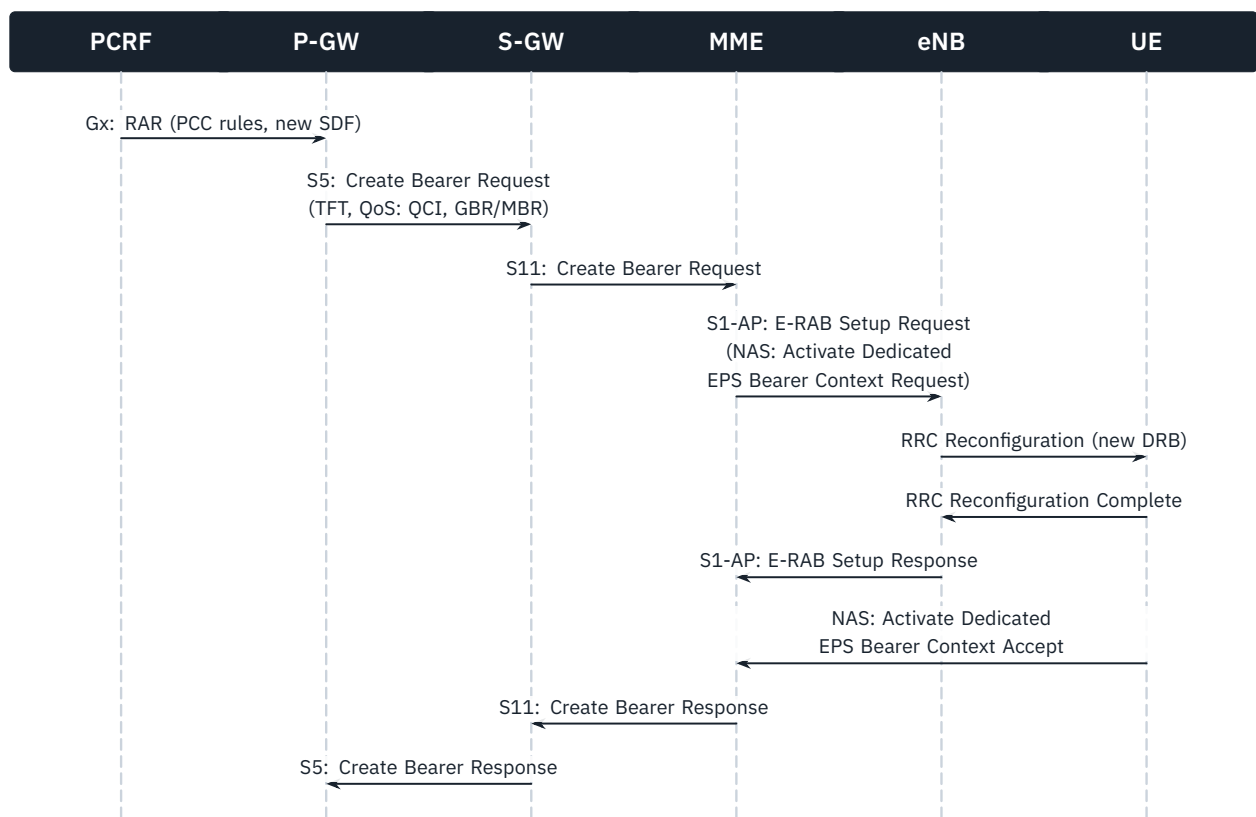


Figure 12.10 Network-initiated dedicated bearer activation

5.3 Dedicated Bearer Modification

- Triggered by PCRF policy change (e.g., QoS upgrade during VoLTE call)
- Uses Update Bearer Request/Response across S5 → S11 → S1-AP

5.4 Dedicated Bearer Deactivation

- Triggered by PCRF (policy removal), P-GW, or MME
- Uses Delete Bearer Request/Response
- Default bearer deactivation = PDN disconnect (all bearers for that PDN removed)

6 MME Selection

6.1 DNS-Based Selection

- eNB resolves MME FQDN from TAI → DNS returns list of MME IPs
- Format: `tac-XXXX.tac-YYYY.mme.epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org`

6.2 Load Balancing

- Each MME advertises a **Relative MME Capacity** (weight, 0-255) via S1 Setup
- eNB distributes new UEs proportionally across pool members
- Higher capacity value = more UEs directed to that MME

6.3 GUMMEI – Globally Unique MME Identifier

$GUMMEI = MCC + MNC + MME \text{ Group ID (MMEGI)} + MME \text{ Code (MMEC)}$

- MMEGI identifies the MME pool
- MMEC identifies a specific MME within the pool
- GUTI = GUMMEI + M-TMSI (temporary UE identity)

7 S1-Flex and MME Pooling

I Concept

- **S1-Flex**: Each eNB connects to **multiple MMEs** in a pool (not just one)
- **MME Pool Area**: Geographic region served by a set of MMEs
- Benefits:
 - **Redundancy**: If one MME fails, others serve UEs
 - **Load sharing**: New UEs distributed by weight
 - **Geo-redundancy**: Pool members can be in different data centers
 - **Graceful scaling**: Add/remove MMEs without service interruption

MME Overload and Rebalancing

- MME sends **Overload Start** to eNBs → eNB redirects new connections to other pool members
- **Rebalancing**: MME requests UEs to re-attach to redistribute load (via implicit detach with re-attach indication)

8 Evolution: MME → AMF (5G)

Aspect	LTE MME	5G AMF
Architecture	Monolithic NE	Microservice (SBI-based)
Protocol	GTPv2-C, Diameter	HTTP/2, JSON (SBI), NGAP
NAS	EMM + ESM (combined)	5G-NAS (MM only; SM separated to SMF)
Session Mgmt	MME handles bearer signaling	Delegated entirely to SMF
Slicing	N/A	Slice-aware, routes to correct SMF per slice
Registration	Attach/TAU	Registration (unified)
State	Stateful	Designed for stateless (UDSF)
Interface to RAN	S1-AP	NGAP (N2)
Interface to UDM	Diameter S6a	Nudm (SBI)

Key Changes:

1. **Session management decoupled**: MME handled both mobility + bearers; in 5G, AMF handles only mobility/security, SMF handles sessions
2. **Service-based interface**: AMF exposes Namf services, consumed by other NFs via HTTP/2
3. **Network slicing**: AMF performs slice selection (NSSF interaction) and routes to slice-specific SMF
4. **Stateless design**: UE context can be stored externally (UDSF) for resilience
5. **Unified registration**: No separate Attach vs TAU — single Registration procedure handles both